

Model Question Paper-II (CBCS Scheme)

USN

Third Semester B.E Degree Examination

AV-MATHEMATICS FOR EC ENGINEERING STREAM (BMATEC301)

TIME: 03 Hours

Max.Marks:100

Note: (i) Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
(ii) Statistical tables and Mathematics Formula handbooks are allowed.

Module -1			M	L	C																
Q.01	a	Find the Fourier series expansion of $f(x)=\begin{cases} x & : 0 \leq x \leq \pi \\ 2\pi-x & : \pi \leq x \leq 2\pi \end{cases}$.	6	L2	CO1																
	b	Expand $f(x)=2x-1$ as a Cosine half range Fourier series in $0 < x < 1$.	7	L2	CO1																
	c	Obtain the constant term and the first coefficients of cosine and sine terms in the Fourier series expansion for the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>9</td><td>18</td><td>24</td><td>28</td><td>26</td><td>20</td></tr></table>	x	0	1	2	3	4	5	y	9	18	24	28	26	20	7	L3	CO1		
x	0	1	2	3	4	5															
y	9	18	24	28	26	20															
OR																					
Q.02	a	Obtain the Fourier series for the triangular wave function $f(x)=\begin{cases} \pi+x, & \text{for } -\pi \leq x \leq 0 \\ \pi-x, & \text{for } 0 \leq x \leq \pi \end{cases}$	6	L2	CO1																
	b	Obtain a half range sine series for $f(x)=(x-1)^2$ in $(0, 1)$	7	L2	CO1																
	c	Obtain the constant term and coefficients of first cosine and sine terms in the expansion of y from the following table <table><tr><td>x</td><td>0</td><td>60°</td><td>120°</td><td>180°</td><td>240°</td><td>300°</td><td>360°</td></tr><tr><td>y</td><td>7.9</td><td>7.2</td><td>3.6</td><td>0.5</td><td>0.9</td><td>6.8</td><td>7.9</td></tr></table>	x	0	60°	120°	180°	240°	300°	360°	y	7.9	7.2	3.6	0.5	0.9	6.8	7.9	7	L3	CO1
x	0	60°	120°	180°	240°	300°	360°														
y	7.9	7.2	3.6	0.5	0.9	6.8	7.9														
Module-2																					
Q. 03	a	Find the Fourier Transform of the function $f(x)=\begin{cases} 1, & \text{for } x \leq a \\ 0, & \text{for } x > a \end{cases}$ Hence evaluate (i) $\int_0^{\infty} \frac{\sin ax \cos sx}{x} dx$, (ii) $\int_0^{\infty} \frac{\sin ax}{x} dx$.	6	L2	CO2																

	b	Find the Fourier cosine Transform of the function $f(x) = e^{-a x }$. Hence evaluate $\int_0^{\infty} \frac{\cos mx}{a^2 + x^2} dx$.	7	L2	CO2
	c	Find the Transform of the sequence $f = [0, 1, 4, 9]^T$ of four values using Fast Fourier Transform.	7	L3	CO2
OR					
Q.04	a	Find the complex Fourier Transform of the function $f(x) = e^{-a^2 x^2}$, $a > 0$	6	L2	CO2
	b	Find the Fourier sine Transform of the function $f(x) = \begin{cases} 4x, & \text{for } 0 < x < 1 \\ 4-x, & \text{for } 1 < x < 4 \\ 0, & \text{for } x > 4 \end{cases}$	7	L1	CO2
	c	Obtain the inverse Fourier cosine Transform of the function $F_c(s) = \frac{\sin as}{s}$, $a > 0$	7	L3	CO2
Module-3					
Q. 05	a	Find the Z-transform of: i) $\sin hn\theta$; ii) $\cos n\theta$	6	L2	CO3
	b	Find the inverse Z-transform of $\frac{z^2 + z}{z^3 + 6z^2 + 11z + 6}$	7	L3	CO3
	c	Solve the difference equation using Z-transform $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$, with $u_0 = 0$ & $u_1 = 1$	7	L3	CO3
OR					
Q. 06	a	Find the Z-transform of $a^n n^2 + 4 \sin\left(\frac{n\pi}{4}\right) + 5$	6	L2	CO3
	b	Find the Inverse Z-transform of $\frac{z^3 - 20z}{(z-2)^3(z-4)}$	7	L3	CO3
	c	If $Z(u_n) = \frac{2z^2 + 3z + 4}{(z-3)^3}$, $ z > 3$. Find u_1, u_2 & u_3	7	L3	CO3
Module-4					
Q. 07	a	Solve $4D^4 - 4D^3 - 23D^2 + 12D + 36 = 0$.	6	L2	CO4
	b	Solve $\frac{d^2 y}{dx^2} - 4y = \cosh(2x-1) + 3^x$.	7	L2	CO4
	c	Solve $x^3 \frac{d^3 y}{dx^3} + 2x^2 \frac{d^2 y}{dx^2} + 2y = 10\left(x + \frac{1}{x}\right)$	7	L3	CO4
OR					
Q. 08	a	Solve $\frac{d^2 y}{dx^2} + 4\frac{dy}{dx} + 4y = 3 \sin x + 4 \cos x$	6	L2	CO4

	b	Solve $(2x+3)^2 \frac{d^2y}{dx^2} - (2x+3) \frac{dy}{dx} - 12y = 6x$.	7	L2	CO4																						
	c	An alternating E.M.F. $E \sin pt$ is applied to a circuit at $t = 0$. Given the equation for the current i as $L \frac{d^2i}{dt^2} + R \frac{di}{dt} + \frac{i}{c} = pE \sin pt$, find i when (i) $CR^2 > 4L$ (ii) $CR^2 < 4L$	7	L3	CO4																						
Module-5																											
Q. 09	a	Find a least square straight line for the following data <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>6</td><td>4</td><td>3</td><td>5</td><td>4</td><td>2</td></tr></table>	x	1	2	3	4	5	6	y	6	4	3	5	4	2	6	L2	CO5								
x	1	2	3	4	5	6																					
y	6	4	3	5	4	2																					
	b	If the coefficient of correlation between the variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}\left(\frac{3}{8}\right)$, show that $\sigma_x = \frac{1}{2}\sigma_y$	7	L2	CO5																						
	c	Determine rank correlation for the following data which shows the marks obtained in two quizzes in mathematics <table><tr><td>Marks in first quiz X</td><td>6</td><td>5</td><td>8</td><td>8</td><td>7</td><td>6</td><td>10</td><td>4</td><td>9</td><td>7</td></tr><tr><td>Marks in second quiz Y</td><td>8</td><td>7</td><td>7</td><td>10</td><td>5</td><td>8</td><td>10</td><td>6</td><td>8</td><td>6</td></tr></table>	Marks in first quiz X	6	5	8	8	7	6	10	4	9	7	Marks in second quiz Y	8	7	7	10	5	8	10	6	8	6	7	L3	CO5
Marks in first quiz X	6	5	8	8	7	6	10	4	9	7																	
Marks in second quiz Y	8	7	7	10	5	8	10	6	8	6																	
OR																											
Q. 10	a	Find a least square quadratic curve for the following data <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>6</td><td>7</td><td>8</td></tr><tr><td>y</td><td>12</td><td>10.5</td><td>10</td><td>8</td><td>7</td><td>2</td><td>8.5</td><td>9</td></tr></table> Estimate y at $x = 6.5$	x	0	1	2	3	4	6	7	8	y	12	10.5	10	8	7	2	8.5	9	6	L2	CO5				
x	0	1	2	3	4	6	7	8																			
y	12	10.5	10	8	7	2	8.5	9																			
	b	For the following data find the correlation coefficient between x and y . <table><tr><td>x</td><td>6</td><td>5</td><td>8</td><td>8</td><td>7</td><td>6</td></tr><tr><td>y</td><td>8</td><td>7</td><td>7</td><td>10</td><td>5</td><td>8</td></tr></table> Also find the standard error estimates.	x	6	5	8	8	7	6	y	8	7	7	10	5	8	7	L3	CO5								
x	6	5	8	8	7	6																					
y	8	7	7	10	5	8																					
	c	Find the two regression lines from the following data <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>14</td><td>12</td><td>16</td><td>11</td><td>14</td><td>12</td></tr></table>	x	1	2	3	4	5	6	y	14	12	16	11	14	12	7	L3	CO5								
x	1	2	3	4	5	6																					
y	14	12	16	11	14	12																					